THE APERIODIC HOLOGRAPHY THEOREM FOR SPECTRE MONOTILES

JAMES F. WALTERS

Abstract. We prove that the sequence of left and right turns defining the boundary of a simply connected patch of Tile(1,1) Spectre monotiles deterministically forces the exact position, orientation, and identity of every tile within the patch's interior.

1. The Topological Constraints

By the Gauss-Bonnet theorem, any simple, closed, counterclockwise loop must exhibit a net turning angle of exactly $+360°$. The boundary of our patch is a sequence of turns drawn from the set $\{L_{90}, L_{60}, S_0, R_{60}, R_{90}\}$. Summing these turns yields the Diophantine Turning Equation:

$$3L_{90} + 2L_{60} - 2R_{60} - 3R_{90} = 12 \tag{1}$$

Furthermore, the edges of the Spectre grid inherently belong to two alternating parity classes. Because $60°$ turns preserve parity and $90°$ turns flip parity, any closed loop must return to its original state, requiring an even number of parity flips. Therefore, we have the parity constraint:

$$(L_{90} + R_{90}) \equiv 0 \pmod 2 \tag{2}$$

2. The Convex Anchor Necessity

We assert that every valid boundary must contain at least one Left $90°$ turn ($L_{90} \geq 1$).

Assume for contradiction that $L_{90} = 0$. By the parity constraint, R_{90} must be an even integer $2k$. Substituting this into the Turning Equation yields:

$$L_{60} - R_{60} - 3k = 6 \tag{3}$$

Geometrically, this implies the macroscopic boundary forms a convex hexagonal hull dominated by Left $60°$ turns, with intermittent Right $90°$ concavities.

However, a single 14-sided Spectre tile contains five $90°$ corners and only one $270°$ corner. In a patch of N tiles, there is a massive surplus of $5N$ orthogonal corners. If $L_{90} = 0$, none of these corners can touch the convex boundary. The limited number of N available $270°$ concavities is vastly insufficient to absorb them. Thus, they must resolve by forming internal "crosses"—vertices where four $90°$ corners meet.

3. The Cyclotomic Integer Lattice Contradiction

To prove internal crosses are geometrically impossible, we map the 12-fold rotational symmetry of the Spectre lattice to the Ring of Cyclotomic Integers, $\mathbb{Z}[\zeta_{12}]$. Every vertex of the tile is represented exactly as a 4-tuple of integers (a, b, c, d).

To rigorously evaluate edge intersections without floating-point approximations, we project these coordinates into the ring $\mathbb{Z}[\sqrt{3}]$ via pure integer scaling, representing every point exactly as $u + v\sqrt{3}$. Evaluating the finite search space of all 625 rotationally locked 4-tile cross combinations using exact algebraic cross-product determinants and winding-number ray-casting reveals a strict geometric contradiction: every single configuration results in the boundary of one tile spilling into the interior region of another.

Because valid simply connected patches cannot contain overlapping tiles, the necessary internal crosses cannot exist. The surplus $90°$ corners cannot be hidden, proving $L_{90} \geq 1$.

4. Chiral Uniqueness and Global Peeling

Because the minimum interior angle of a Spectre tile is 90°, it is impossible for two tiles to combine to form a 90° convex patch corner. Therefore, exactly one tile occupies every Left 90° boundary turn.

The Spectre tile is strictly chiral. An evaluation of its 14-turn perimeter confirms that the local neighborhoods flanking its five 90° corners are geometrically distinct. Reading any 3-turn sequence on the boundary centered on a Left 90° turn uniquely locks the exact orientation and position of that specific tile.

Having uniquely identified a boundary tile, we recursively "peel" it away by removing its exterior edges from the boundary path and splicing in the inverse of its interior edges. This list surgery strictly preserves the net 360° curvature and directional consistency of the macroscopic loop while reducing the patch area by one. By continuously identifying Left 90° anchors and peeling, the boundary deterministically collapses, holographically decoding the entire interior. ■